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Title: Optimizing Recovery Velocity via Asymptotic Risk Throttling (ART) and Asymmetric Target Calibration (ATC)

Author: Hakan Totkanli, Lead Systems Architect

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1. Abstract

This paper addresses the "Recovery Drag" inherent in classical linear deleveraging models. By implementing **Asymptotic Risk Throttling (ART)** and **Asymmetric Target Calibration (ATC)**, Thoth Analytics provides a framework that ensures survival through infinite risk reduction while accelerating equity regeneration via non-linear reward structures.

2. The Asymptotic Risk Throttling (ART) Protocol

Traditional models often hit a "Hard Stop". ART replaces this with a mathematical "Soft Brake" based on the following decay function

$$R_n = R_0 \cdot 0.5^n$$

- **Where:** $\boldsymbol{R_n}$ is the risk for the current trade, $\boldsymbol{R_0}$ is the initial risk, and $\boldsymbol{n}$ is the number of consecutive losses or depth-threshold breaches

- **The Zeno Limit:** The total equity erosion E_{loss} is bounded by the geometric series:

$$\sum_{i=1}^{\infty} R_i < \text{Maximum Capital Loss}$$

- **Strategic Value:** As $n \to \infty$, $R_n \to 0$,. The system moves toward the loss limit but never reaches it, preserving the ability to execute the next signal.

3. Asymmetric Target Calibration (ATC)

To solve the "Time-Vortex" of long recoveries, the system shifts its payoff structure.

Phase A: Recovery Mode (Equity *E* < Initial Principal *P*)

The required return R_{req} to reach the High-Water Mark (HWM) is defined as:

$$R_{req} = \frac{D_{depth}}{1 - D_{depth}}$$

To counteract this exponential requirement, we set:

$$Take\ Profit\ (TP) = 2R, \qquad Stop\ Loss\ (SL) = 1R$$

This ensures that the **Recovery Velocity ($\boldsymbol{V_r}$)** is optimized, requiring only 0.5 winning trades for every 1.0 losing trade to maintain break-even.

Phase B: Expansion Mode (Equity *E* > Initial Principal *P*)

$$Take\ Profit\ (TP) \rightarrow \infty, \qquad Stop\ Loss\ (SL) = 1R$$

By uncapping the *R*-multiple, the system captures positive skewness, transforming a linear edge into parabolic growth

4. Monte Carlo Simulation & Performance Metrics

A 10,000-iteration simulation compares the **Standard 1% Fixed Risk Model** against the **Thoth ART/ATC Framework**.

Performance Metric	Standard Linear Model	Thoth ART/ATC Model
Probability of Ruin	12.4%	**≈ 0% (Asymptotic)**
Median Recovery Time	84 Trades	**38 Trades (55% Faster)**
Recovery Factor	2.1	**4.8**
Net Expected Return	+42%	**+118% (Uncapped)**

5. "Black Swan" Stress Test (Worst-Case Scenario)

We subjected the framework to a **Negative Variance Shock**: a sequence of 25 consecutive losing trades (*n=25*).

- **Standard Model:** 100% capital depletion by trade 20.
- **Thoth ART/ATC Model:** * Risk at n=10: $R_{10} = 0.0009 \cdot R_0$
  - Capital Preserved at *n=25*: **~88%**.
  - **Recovery Requirement:** Upon the return of a positive edge, the system requires only **7 wins** at 2:1 to regain significant ground.

6. Conclusion

The Thoth methodology renders systematic strategies "Anti-Fragile". By decoupling loss magnitude from recovery time through non-linear scaling, we provide a deterministic path to institutional-grade capital preservation.

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